Topic 4 - General Vector Spaces

4.1 General vector spaces

Fields

The *scalars* are members of a number system $\mathbb{F}$ called a **field** where addition, multiplication, subtraction and division by a non-zero scalar are defined.

Every field $\mathbb{F}$ contains:
1. A zero scalar s.t. $\alpha + 0 = \alpha$
2. A unit scalar (with $1 \neq 0$) s.t. $1\alpha = \alpha$

> **Note** — **Subject scope**
>
> MAST10007 mostly looks at $\mathbb{F} = \mathbb{R}$ and $\mathbb{F} = \mathbb{C}$, but also looks at $\mathbb{F} = \mathbb{F}_2 = \{0, 1\}$, the integers modulo 2.

Vector spaces

The aim of vector spaces is to extend the properties of vectors from $\mathbb{R}^n$ to more general sets of objects.

A **vector space** is a set V with two operations defined:
1. vector addition
2. scalar multiplication

We want these two to satisfy the basic properties of vectors in $\mathbb{R}^n$, which leads to a list of 10 *axioms* that give the definition of a vector space.

> **Definition** — **Vector space**
>
> For a fixed field of scalars $\mathbb{F}$, a **vector space** *over* F is a non-empty set V with two operations: vector addition and scalar multiplication. The operations are required to satisfy 10 axioms:
>
> > **Axiom** — **Additive axioms**
> >
> > Vector addition behaves well:
> > 1. $\boldsymbol{u} + \boldsymbol{v} \in V$ (closure in vector addition)
> > 2. $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$ (associativity)
> > 3. $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ (commutativity)
> >
> > There must be a zero and an inverse:
> > 1. There exists a vector $\mathbf{0} \in V$ s.t. $\boldsymbol{v} + \mathbf{0} = \boldsymbol{v}$ for all $\boldsymbol{v} \in V$ (existence of zero vector)
> > 2. For all $\boldsymbol{v} \in V$, there exists a vector $-\boldsymbol{v}$ s.t. $\boldsymbol{v} + (-\boldsymbol{v}) = \mathbf{0}$ (additive inverse)
>
> > **Axiom** — **Multiplicative axioms**
> >
> > Scalar multiplication behaves well:
> > 1. $c\boldsymbol{v} \in V$ (closure under scalar multiplication)
> > 2. $c(\boldsymbol{u} + \boldsymbol{v}) = c\boldsymbol{u} + c\boldsymbol{v}$ (distributivity over vector addition)

3. $(c+d)\boldsymbol{v} = c\boldsymbol{v} + d\boldsymbol{v}$ (distributivity over scalar addition)
4. $(cd)\boldsymbol{v} = c(d\boldsymbol{v})$ (compatibility of scalar multiplication)
5. $1\boldsymbol{v} = \boldsymbol{v}$ (scalar multiplicative identity)

Examples of vector spaces

Example **Complex numbers $\mathbb{C}$**

$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ forms a vector space over $\mathbb{R}$, with addition

$$(a+bi) + (c+di) = (a+c) + (b+d)i$$

and scalar multiplication $\lambda(a+bi) = \lambda a + \lambda bi$ for $\lambda \in \mathbb{R}$. The zero vector is $0 + 0i$.

Example **Integers mod 2: $\mathbb{Z}_2^n$**

$\mathbb{Z}_2 = \{0, 1\}$ with arithmetic mod 2 is a field, and $\mathbb{Z}_2^n$ forms a vector space over $\mathbb{Z}_2$. Addition is bitwise XOR and scalar multiplication is trivial ($0 \cdot \boldsymbol{v} = \mathbf{0}$, $1 \cdot \boldsymbol{v} = \boldsymbol{v}$). This space is fundamental in coding theory and cryptography.

Example **Polynomials $\mathbb{P}_n$**

The set of all polynomials of degree at most n,

$$\mathbb{P}_n = \{a_0 + a_1 x + \ldots + a_n x^n \mid a_i \in \mathbb{R}\}$$

forms a vector space over $\mathbb{R}$ under standard polynomial addition and scalar multiplication. The zero vector is the zero polynomial. More generally, $\mathbb{P}$ (polynomials of arbitrary degree) is also a vector space.

Example **Function spaces $\mathcal{F}(X, \mathbb{R})$**

The set of all functions $f : X \to \mathbb{R}$ from a non-empty set X forms a vector space over $\mathbb{R}$, with pointwise operations

$$(f+g)(x) = f(x) + g(x), \quad (cf)(x) = c \cdot f(x)$$

The zero vector is the constant zero function $\mathbf{0}(x) = 0$. Important subspaces include:
- $C[a, b]$: continuous functions on $[a, b]$
- $C^k[a, b]$: k-times continuously differentiable functions
- $C^\infty[a, b]$: smooth (infinitely differentiable) functions

4.2 Subspaces

Definition **Subspace**

A **subspace** W of a vector space V is a subset $W \subseteq V$ that is a vector space using the vector addition and scalar multiplaction definitions of V.

The subspace theorem can be used to check if a vector space is a subspace (and hence a vector space) without checking all 10 axioms.

Theorem **Subspace theorem**

For a vector space V with scalars $\mathbb{F}$, A subset $W \subseteq V$ is a subspace of V iff:
1. W contains the zero vector $\mathbf{0}$ from V
2. W is closed under vector addition ($\boldsymbol{u}, \boldsymbol{v} \in W \Rightarrow \boldsymbol{u} + \boldsymbol{v} \in W$)
3. W is closed under scalar multiplication ($\boldsymbol{u}, \boldsymbol{v} \in W \Rightarrow \boldsymbol{u}\boldsymbol{v} \in W$)

Example **A plane through the origin in $\mathbb{R}^3$**

Let $W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x - y + z = 0\}$. We claim W is a subspace of $\mathbb{R}^3$.

Zero: $2(0) - 0 + 0 = 0$ ✓

Closure under addition: Let $\boldsymbol{u} = (u_1, u_2, u_3), \boldsymbol{v} = (v_1, v_2, v_3) \in W$, so

$$2u_1 - u_2 + u_3 = 0 \quad \text{and} \quad 2v_1 - v_2 + v_3 = 0.$$

Adding these equations gives $2(u_1 + v_1) - (u_2 + v_2) + (u_3 + v_3) = 0$, so $\boldsymbol{u} + \boldsymbol{v} \in W$. ✓

Closure under scalar multiplication: Let $\boldsymbol{v} = (v_1, v_2, v_3) \in W$ and $c \in \mathbb{R}$. Then

$$2(cv_1) - (cv_2) + (cv_3) = c(2v_1 - v_2 + v_3) = c \cdot 0 = 0$$

so $c\boldsymbol{v} \in W$. ✓

Hence W is a subspace of $\mathbb{R}^3$.

Example **Symmetric matrices S_n**

Let $S_n = \{A \in M_{n\times n}(\mathbb{R}) \mid A^{\mathbb{T}} = A\}$ be the set of $n \times n$ symmetric matrices. We claim S_n is a subspace of $M_{n\times n}(\mathbb{R})$, where the vector space operations are entrywise addition and scalar multiplication.

Zero: The zero matrix $\mathbf{0}$ satisfies $\mathbf{0}^{\mathbb{T}} = \mathbf{0}$, so $\mathbf{0} \in S_n$. ✓

Closure under addition: Let $A, B \in S_n$, so $A^{\mathbb{T}} = A$ and $B^{\mathbb{T}} = B$. Using properties of the transpose,

$$(A + B)^{\mathbb{T}} = A^{\mathbb{T}} + B^{\mathbb{T}} = A + B$$

so $A + B \in S_n$. ✓

Closure under scalar multiplication: Let $A \in S_n$ and $c \in \mathbb{R}$. Then

$$(cA)^{\mathbb{T}} = cA^{\mathbb{T}} = cA$$

so $cA \in S_n$. ✓

Hence S_n is a subspace of $M_{n\times n}(\mathbb{R})$.

> **Remark**
>
> The dimension of S_n is $\binom{n+1}{2} = \frac{n(n+1)}{2}$, since a symmetric matrix is uniquely determined by its entries on and above the main diagonal.

4.3 Linear combinations and spans

Definition **Linear combination**

A **linear combination** of vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_k \in V$ is the sum of scalar multiples of the vectors:

$$\sum_{i=1}^{k} \alpha_i \boldsymbol{v}_i = \alpha_1 \boldsymbol{v}_1, \alpha_2 \boldsymbol{v}_2, ..., \alpha_k \boldsymbol{v}_k$$

where each $\alpha_i \in \mathbb{F}$.

Definition **Span of a set of vectors**

For a vector space V over $\mathbb{F}$, the **span** of vectors $\boldsymbol{v}_1 ... \boldsymbol{v}_k \in V$ is the *set of all linear combinations* of the vectors:

$$\text{span}\{\boldsymbol{v}_1, ..., \boldsymbol{v}_k\} = \{\alpha_1 \boldsymbol{v}_1 + ... + \alpha_k \boldsymbol{v}_k \mid \alpha_1, ..., \alpha_k \in \mathbb{F}\}$$

Theorem **Spans and subspaces**

For vectors in a vector space, the span of these vectors is a subspace of the vector space.

General method to determine if vectors span $\mathbb{F}^n$

If $A = [\boldsymbol{v}_1 ... \boldsymbol{v}_k]$ where $\boldsymbol{v}_i \in \mathbb{F}^n$ are column vectors, then:

1. $\boldsymbol{v}_1, ..., \boldsymbol{v}_k$ span $\mathbb{F}^n \Leftrightarrow$ the linear system with augmented matrix $[A \mid \boldsymbol{w}]$ has a solution for all $\boldsymbol{w} \in \mathbb{F}^n$
2. $\boldsymbol{v}_1, ..., \boldsymbol{v}_k$ span $\mathbb{F}^n \Leftrightarrow$ rank(A) = n

Theorem **rank and span**

If $k < n$, then vectors $\boldsymbol{v}_1, ..., \boldsymbol{v}_k \in \mathbb{F}^n$ do not span $\mathbb{F}^n$.

Proof: Since $A = [\boldsymbol{v}_1 ... \boldsymbol{v}_k]$ has k columns, $\text{rank}(A) \leq k$. Since $k < n$ this implies that $\text{rank}(A) \neq n$. Hence, $\boldsymbol{v}_1, ..., \boldsymbol{v}_k$ do not span $\mathbb{F}^n$ □

4.4 Linear dependence and linear independence

A vector space can have many sets that span it. To get a small spanning set, we need to remove 'redundant' vectors. This leads to the ideas of **linear dependence and independence**.

Definition **Linear dependence**

The vectors $\boldsymbol{v}_1, ..., \boldsymbol{v}_k \in V$ are **linearly dependent** if there exist scalars $\alpha_1, ..., \alpha_k \in \mathbb{F}$, *with at least one α_i non-zero*, s.t.

$$\sum_{i=1}^{k} \alpha_i \boldsymbol{v}_i = \boldsymbol{0}$$

.

Definition **Linear independence**

The vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in V$ are **linearly independent** if there exist scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, s.t.

$$\sum_{i=1}^{k} \alpha_i \boldsymbol{v}_i = \mathbf{0}$$

has only the solution $\alpha_i = 0$ for all $i < k$.

Theorem **Linear dependence and combinations**

The vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k (k \geq 2)$ are *linearly dependent* iff one vector is a linear combination of the others.

Example **Vectors in RR^3**

Question. Are the vectors

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

linearly independent in $\mathbb{R}^3$?

Solution. Row-reduce the matrix with these as rows:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} R_3 \leftarrow R_3 - R_1$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} R_3 \leftarrow R_3 - R_2$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{pmatrix}.$$

Three pivots, so $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are **linearly independent**.

Example **Functions in $C(\mathbb{R})$**

Question. Are $f_1(x) = \sin^2 x$, $f_2(x) = \cos^2 x$, $f_3(x) = \cos(2x)$ linearly independent in $C(\mathbb{R})$?

Solution. Suppose $\alpha f_1 + \beta f_2 + \gamma f_3 = 0$ for all x. Evaluating at $x = 0, \frac{\pi}{2}, \frac{\pi}{4}$ gives the system

$$\beta + \gamma = 0, \quad \alpha - \gamma = 0, \quad \alpha + \beta = 0,$$

which has the non-trivial solution $(\alpha, \beta, \gamma) = (1, -1, 1)$. So f_1, f_2, f_3 are **linearly dependent**,.

General method for determining if vectors are linearly independent

If $A = [\boldsymbol{v}_1 \ldots \boldsymbol{v}_k]$ where $\boldsymbol{v}_i, \ldots, \boldsymbol{v}_k \in \mathbb{F}^n$ are column vectors, then:

1. $\boldsymbol{v}_i, \ldots, \boldsymbol{v}_k \in \mathbb{F}^n$ are linearly dependent $\Leftrightarrow$ the linear system with augmented matrix $[A \mid \mathbf{0}]$ has a unique solution.
2. $\boldsymbol{v}_i, \ldots, \boldsymbol{v}_k \in \mathbb{F}^n$ are linearly independent $\Leftrightarrow$ $\operatorname{rank}(A) = k$.
3. If $k = n$ then

$$\boldsymbol{v}_i, \ldots, \boldsymbol{v}_k \in \mathbb{F}^n \text{ are linearly independent} \Leftrightarrow \det(A) \neq 0$$

Theorem **Vector count**

Let $\boldsymbol{v}_i, \ldots, \boldsymbol{v}_k \in \mathbb{F}^n$. If $k > n$, then the vectors are linearly dependent.

Proof: Let $A = [\boldsymbol{v}_i \ldots \boldsymbol{v}_k]$ where $\boldsymbol{v}_i \in \mathbb{F}^n$. Then $\operatorname{rank}(A) \leq n$. Since $k < n$ this implies that rank $(A) \neq k$. Hence the vectors are linearly dependent. □

Linear combinations via reduced row-echelon form

Let $A = [\boldsymbol{v}_i \ldots \boldsymbol{v}_k]$ and $B = [\boldsymbol{u}_i \ldots \boldsymbol{u}_k]$ be matrices with vectors in columns.

If A and B are row-equivalent (ie. $A \sim B$), then the columns of A satisfy the *same linear relations* as the columns of B:

$$\alpha_1 \boldsymbol{v}_1 + \ldots + \alpha_k \boldsymbol{v}_k = 0 \Leftrightarrow \alpha_1 \boldsymbol{u}_1 + \ldots + \alpha_k \boldsymbol{u}_k = 0$$

If $A \sim B$ and B is in RREF, then the relations between the $\boldsymbol{v}$ vectors can be read off from B.

Example **Row-equivalent matrices share column relations**

Let $A = [\boldsymbol{v}_1 \mid \boldsymbol{v}_2 \mid \boldsymbol{v}_3 \mid \boldsymbol{v}_4]$ where

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_4 = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}.$$

Row-reducing A to RREF gives $A \sim B$:

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 1 & 1 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = B.$$

Columns 1 and 2 are pivot columns; columns 3 and 4 are free. Solving $B\boldsymbol{x} = \mathbf{0}$ gives $x_1 = x_3 - x_4$ and $x_2 = -x_3 - x_4$, so the null space is spanned by

$$\boldsymbol{x} = x_3 \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

These two vectors give the **complete** set of linear relations among the columns. By row-equivalence the same relations hold for A:

$$\boldsymbol{v}_1 - \boldsymbol{v}_2 + \boldsymbol{v}_3 = \mathbf{0}, \quad -\boldsymbol{v}_1 - \boldsymbol{v}_2 + \boldsymbol{v}_4 = \mathbf{0}.$$

We can verify the first directly: $(1, 2, 1) - (2, 3, 1) + (1, 1, 0) = (0, 0, 0)$. □

4.5 Bases and Dimension

Sometimes spanning sets are larger than they need to be, so to get an efficient description of a vector space V, we want to find a set with the *minimum number of vectors* needed to span V.

Definition — **Basis**

A **basis** for a vector space V is a set of vectors in V which:
1. Spans V, and
2. Is linearly independent.

Note — **Unique bases**

A vector space V can have many different bases, they are not unique, eg. any two linearly independent vectors in $\mathbb{R}^2$ will span $\mathbb{R}^2$

Theorem — **Existence of bases**

Let V be a vector space. Then
1. V has a basis.
2. Any set that spans V contains a basis for V
3. Any linearly independent set in V can be extended to a basis of V

Theorem — **Length of basis**

For a vector space V with basis {bold(v)*1, ..., bold(v)*k}:
1. A subset of V with more than k vectors is linearly dependent
2. A subset of V with fewer than k vectors does not span V
3. Any two bases of V have the same number of elements, k

This allows us to define the dimension of a vector space.

Definition — **Dimension**

The **dimension** of a vector space V is the number of vectors in any basis for V. This is denoted by $\dim(V)$.

We call V **finite dimensional** if it has a basis with a finite number of elements, and infinite dimensional otherwise.

Note — **Empty basis**

If $V = \{\mathbf{0}\}$, we say that $\dim(V) = 0$ and a basis for V is the empty set.

Standard bases

- The **standard basis for $\mathbb{R}^n$** is

$$\begin{gathered}\{(1,0,0,\ldots,0),\\ (0,1,0,\ldots,0),\\ \ldots,\\ (0,0,0,\ldots,1)\}.\end{gathered}$$

  These vectors are typically denoted as $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$. (In $\mathbb{R}^3$ the standard basis is also denoted by $\{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$)

  The dimension of $\mathbb{R}^n$ is n.

- The **standard basis for $M_{m,n}$** is

$$\left\{\begin{pmatrix}1 & 0 & \ldots & 0\\ 0 & 0 & \ldots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \ldots & 0\end{pmatrix}, \begin{pmatrix}0 & 1 & \ldots & 0\\ 0 & 0 & \ldots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \ldots & 0\end{pmatrix}, \ldots, \begin{pmatrix}0 & 0 & \ldots & 0\\ 0 & 0 & \ldots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \ldots & 1\end{pmatrix}\right\}.$$

  The dimension of $M_{m,n}$ is mn.

- The **standard basis for $\mathcal{P}_n$** is

$$\{1, x, x^2, \ldots, x^n\}.$$

  The dimension of $\mathcal{P}_n$ is $n+1$.

- Let $\mathcal{P}$ be the set of **all** polynomials of all possible degrees:

$$\begin{gathered}\mathcal{P} = \{a_0 + a_1 x + \ldots + a_n x^n \mid n \in \mathbb{N},\\ a_0, a_1, \ldots, a_n \in \mathbb{R}\}.\end{gathered}$$

  where $\mathbb{N}$ denotes the set of natural numbers. $\mathcal{P}$ is an ***infinite dimensional*** vector space. The **standard basis for $\mathcal{P}$** is

$$\{1, x, x^2, \ldots\}.$$

Theorem

Let V be a vector space, and suppose $\dim(V) = n$ is finite.

1. If a set with exactly n elements spans V, then it is a basis of V
2. If a linearly independent subset of V has exactly n elements, then it is a basis for V

Finding a basis for the span of a set of vectors in $\mathbb{F}^n$

Recall that if $A \sim B$, then the columns of A and B satisfy the same linear relationships.

Method 1: Using columns

To find a basis for $V = \text{span}\{\boldsymbol{v}_1, ..., \boldsymbol{v}_k\}$ where $\boldsymbol{v}_1 ... \boldsymbol{v}_k \in \mathbb{F}^n$.

1. Form the matrix A with *columns* $\boldsymbol{v}_1, ..., \boldsymbol{v}_k$.
2. Reduce A to row-echelon form B.
3. The columns of A corresponding to the leading entries of B give a basis for V.

> **Note**
>
> This method gives a basis that *is a subset* of the original set of vectors.

Method 2: Using rows

To find a basis for $V = \text{span}\{\boldsymbol{v}_1, ..., \boldsymbol{v}_k\}$ where $\boldsymbol{v}_1 ... \boldsymbol{v}_k \in \mathbb{F}^n$.

1. Form the matrix A with *rows* $\boldsymbol{v}_1, ..., \boldsymbol{v}_k$.
2. Reduce A to row-echelon form B.
3. The non-zero rows of B give a basis for V.

> **Note**
>
> This method gives a basis that *usually is not a subset* of the original set of vectors.
>
> This works because elementary row operations do not change the subspace spanned by the rows, and in the row-echelon form the non-zero rows are linearly independent.

4.6 Solution space, column space, row space